

SEIS 606 Homework 4: Testing, Testing...

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Report on Week 3 Testing

As of week 3's class, the app might as well not have existed. The web page was blank and despite the large amount of tokens Copilot used, it failed to generate anything tangible. I will have to be more careful about reviewing output in future iterations. When testing my small group's apps, I also saw how much the first screen matters. A feature does not count for much if you cannot tell whether it loaded or what it is showing. This time I ran the app and checked the page itself, not just whether code had been generated. The blank page is gone, but it was a good reminder to verify what a user actually sees.

Recent Modifications

The difference since week 3 is night and day. The dashboard works, looks how I want it to, and is relatively easy to add features to (i.e. maintainable).

Unfortunately for the purposes of HW4, I didn't make dramatic changes to the spec to get past that hurdle. All I did was acknowledge that *Streamlit* is probably not the right tool for the job, and instead asked for standard HTML / CSS / JS for the web UI. I kept Python as the backend for handling API requests from the UI, and for data aggregation from the servers. After that, Copilot worked its magic a whole lot better than it did before.

With the exception of the inventory management feature, I think the web UI is close to what I envisioned originally. I look forward to what more I can refine before the project's due date.

Dashboard Screenshots

These are screenshots of the running dashboard. Making this site publicly accessible sounds like a security nightmare given the level of access the backend has into my network, so it's local-only for now. I utilized AI to generate accessibility captions for each image.

Dashboard Overview

As seen on my 16:9 1440p display:

The site resizes for mobile devices as well:

Dynamic Cluster Topology

This panel displays the topology of my PVE nodes. Each node's containers are visible below them. When a node is connected, its dotted line will turn thicker and sport the associated color.

VM and Container View

This section allows me to filter and search through my virtual machines and containers. By default, it shows all guests across all nodes. Filters allow me to quickly locate specific guests and view their status across different nodes.

Network DHCP

This panel shows the DHCP leases for devices on my local network. Below is a clip of the guests on my IoT and Guest VLANs. I filter the lists between two different panels.

Node and Hardware Health

I tried to make these system summary cards dense yet easy to read at a glance. Each server's CPU usage, RAM usage, and ZFS pool usage are displayed as prominent circles / meters. Specific values for RAM and ZFS ARC (also RAM) usage are shown to the right. Underneath are colored cards that highlight hardware errors, SMART status, and other critical alerts. These were inspired by me fighting every server except one over the last week and a half.



Figure 1: Full dashboard overview with node metrics, LAN devices discovered from router leases, topology, guests, and haiku

My main server, Kex, suffered hardware failures and I wanted to proactively catch them next time. Clicking on the *HW Errors* badge opens a detailed view of recent incidents in a modal. The background is blurred for greater visual separation.

Scrolling down gives you a table of incidents. Summaries are useful, but knowing specific errors is crucial for debugging.

Clicking on the *SMART* badge opens a modal displaying the health status of my storage devices, in addition to the pools they belong. Funny enough, this identified overheating drives that I wouldn't have noticed otherwise.

Activity Logging

This modal displays recent web UI activity, including requests made to the dashboard and actions taken by users. I figured repeating the nodes' logs themselves would be redundant, though that might be a feature in a future version.

Dashboard Styling

Originally, this card was here to fill in blank space. Now, that's less of an issue, but I decided to keep it anyway. The world needs more whimsy and I'll start by adding that to my dashboard.

The UI's header used to have a second line with a silly slogan. I've since consolidated that header element, but the slogan went too hard to throw away forever. I moved it to the bottom-right corner, where my favorite command-line utility *cowsay* now displays it. It can be seen below.

I included more than just the cow in this screenshot to highlight the background details and Frutiger Aero-inspired bubble graphics.



Figure 2: Mobile version of the dashboard overview with node metrics, LAN devices discovered from router leases, topology, guests, and haiku

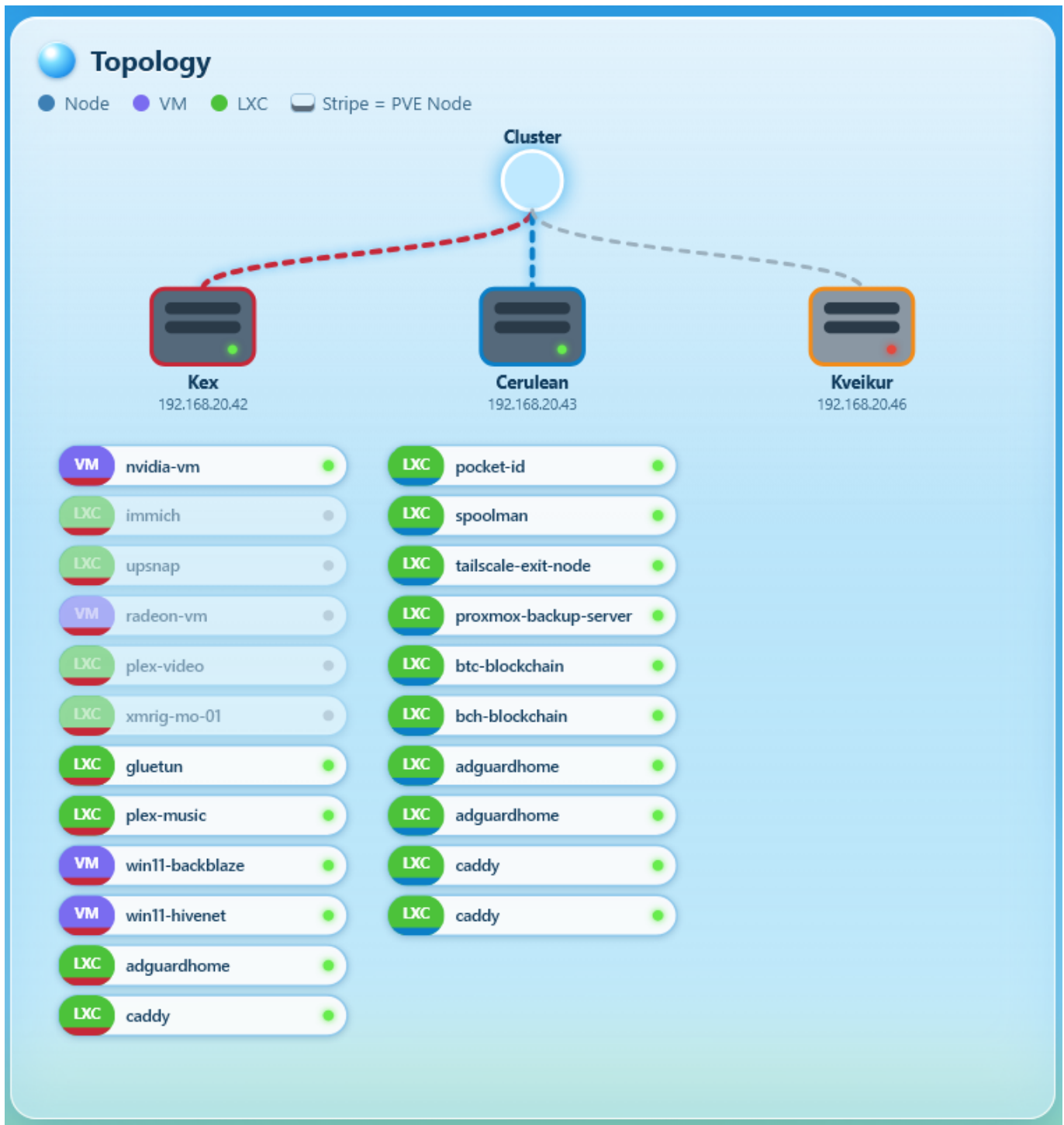


Figure 3: Cluster topology showing nodes and their guests



Figure 4: VM filter selected for Kex, showing running and stopped virtual machines

Guest / IoT Networks

OpenWRT · 192.168.10.1

KP125M

IPv4

172.16.0.206

MAC

98:BA:5F:61:F5:4E

ecoflow

IPv4

172.16.0.238

MAC

9C:9E:6E:75:BE:CC

nerd-qaxe

IPv4

172.16.10.167

MAC

80:B5:4E:D7:35:34

りんご-m4

IPv4

172.16.20.4

MAC

84:2F:57:AB:BD:18

172.16.0.82

IPv4

172.16.0.82

MAC

F0:09:0D:C9:2F:C3

Figure 5: Searchable guest list with node and status filters

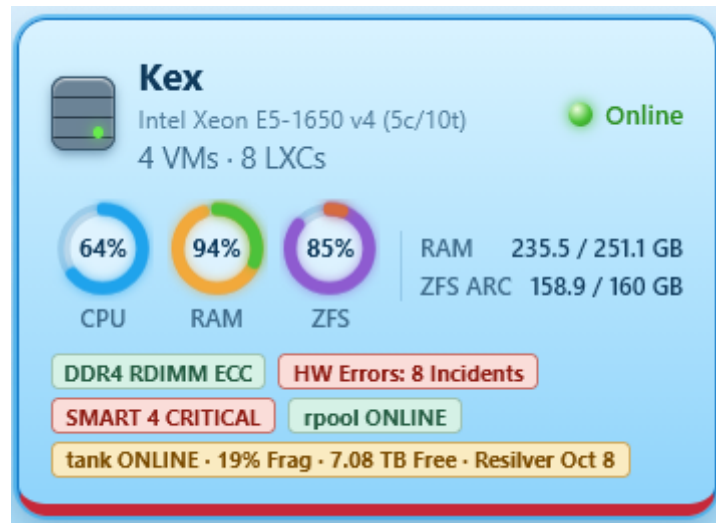


Figure 6: Kex system card with CPU, RAM, ZFS, SMART, and hardware error summaries



Figure 7: Hardware error summary and ECC counters for Kex

Incidents					
Scope	Class	Evidence	Events	First seen	Last seen
This boot	ECC memory · corrected	Repeated corrected ECC memory errors seen on 6 boots ► Raw records (latest 10 of 11)	11 (1 in 1 h, 11 in 24 h)	10/8/2026, 12:31:20 AM	10/8/2026, 8:15:55 PM
This boot	CPU machine check · unspecified	CMCI storm reported by the CPU machine-check subsystem seen on 6 boots · note=CMCI storm ► Raw records (latest 10 of 35)	35 (2 in 1 h, 35 in 24 h)	10/8/2026, 12:31:17 AM	10/8/2026, 8:15:55 PM
This boot	ECC memory · corrected	Repeated corrected ECC errors at the same memory location (CPU_SrcID#0_Ha#0_Chan#0_DIMM#0), address 0x17c08ac000 same DIMM CPU_SrcID#0_Ha#0_Chan#0_DIMM#0 ×4 · same address 0x17c08ac000 ×4 · same page frame 0x17c08ac ×4 · seen on 2 boots · dimm=CPU_SrcID#0_Ha#0_Chan#0_DIMM#0 · address=0x17c08ac000 · page_frame=0x17c08ac · controller=mc0 ► Raw records (latest 4 of 4)	4 (1 in 1 h, 4 in 24 h)	10/8/2026, 5:39:55 AM	10/8/2026, 8:15:55 PM
This boot	Page soft-offline · unspecified	Kernel soft-offlined memory page 0x17c08ac (page retirement is containment, not a repair) same page frame 0x17c08ac ×8 · seen on 3 boots · page_frame=0x17c08ac ► Raw records (latest 8 of 8)	8 (1 in 1 h, 8 in 24 h)	10/8/2026, 3:10:59 AM	10/8/2026, 8:15:55 PM
This boot	ECC memory · corrected	Corrected ECC memory error on CPU_SrcID#0_Ha#0_Chan#0_DIMM#1 seen on 3 boots · dimm=CPU_SrcID#0_Ha#0_Chan#0_DIMM#1 · address=0x17c08aca80 · page_frame=0x17c08ac · controller=mc0 ► Raw records (latest 1 of 1)	1 (0 in 1 h, 1 in 24 h)	10/8/2026, 3:10:59 AM	10/8/2026, 3:10:59 AM
This boot	CPU machine check · corrected	Corrected internal parity error (bank 0) seen on 4 boots · bank=0 · type=internal parity error ► Raw records (latest 1 of 1)	1 (0 in 1 h, 1 in 24 h)	10/8/2026, 12:35:39 AM	10/8/2026, 12:35:39 AM
This boot	ECC memory · unspecified	Repeated ECC memory errors seen on 6 boots ► Raw records (latest 9 of 9)	9 (0 in 1 h, 9 in 24 h)	10/8/2026, 12:31:20 AM	10/8/2026, 12:35:39 AM
This boot	Storage path · unspecified	Repeated storage-path errors seen on 5 boots ► Raw records (latest 4 of 4)	4 (0 in 1 h, 4 in 24 h)	10/8/2026, 12:31:17 AM	10/8/2026, 12:31:17 AM

Figure 8: Detailed hardware error incidents with evidence and timestamps

Storage Health · Kex

Disks (SMART)

Status	Device	Model	Serial #	Type	Temp	Age	Findings
OK	/dev/sda	TOSHIBA PX02SMF080	1550A0EST27A	SSD	56 °C	7y 252d	Normal
Critical	/dev/sdb	ST26000DM000-3Y8103	ZXA0Y6RH	HDD	50 °C	1y 24d	Temperature 50 °C (critical at 50 °C)
Critical	/dev/sdc	ST26000DM000-3Y8103	ZXA0Q3GR	HDD	50 °C	1y 24d	Temperature 50 °C (critical at 50 °C)
OK	/dev/sdd	TOSHIBA PX02SMF080	1550A0CVT27A	SSD	59 °C	7y 254d	Normal
Warning	/dev/sde	ST26000DM000-3Y8103	ZXA0BEA9	HDD	48 °C	1y 24d	Temperature 48 °C (warning at 45 °C)
Warning	/dev/sdf	ST26000DM000-3Y8103	ZXA0M25L	HDD	45 °C	1y 24d	Temperature 45 °C (warning at 45 °C)
OK	/dev/sdg	TOSHIBA PX02SMF080	1550A0BCT27A	SSD	51 °C	7y 254d	Normal
OK	/dev/sdh	TOSHIBA PX02SMF080	1550A0ALT27A	SSD	54 °C	7y 253d	Normal
OK	/dev/nvme0	XPB GAMMIX S11 Pro	2K0320080209	NVME	44 °C	2y 294d	Normal
OK	/dev/nvme1	T-FORCE TM8FP8002T	112104260170621	NVME	40 °C	1y 135d	Normal

ZFS Pools

Status	Pool	Capacity	Used / Size	Free	Frag	Layout	Last Scan	Findings
ONLINE	rpool	18% Full	352.9 GB / 1.86 TB	1.51 TB	36%	1 × Mirror (2 Disks per Mirror)	Resilver Oct 3	Normal
ONLINE	tank	85% Full	41.65 TB / 48.73 TB	7.08 TB	19%	2 × Mirror (2 Disks per Mirror)	Resilver Oct 8	85% full

Figure 9: Storage health modal showing SMART disk status and ZFS pool details

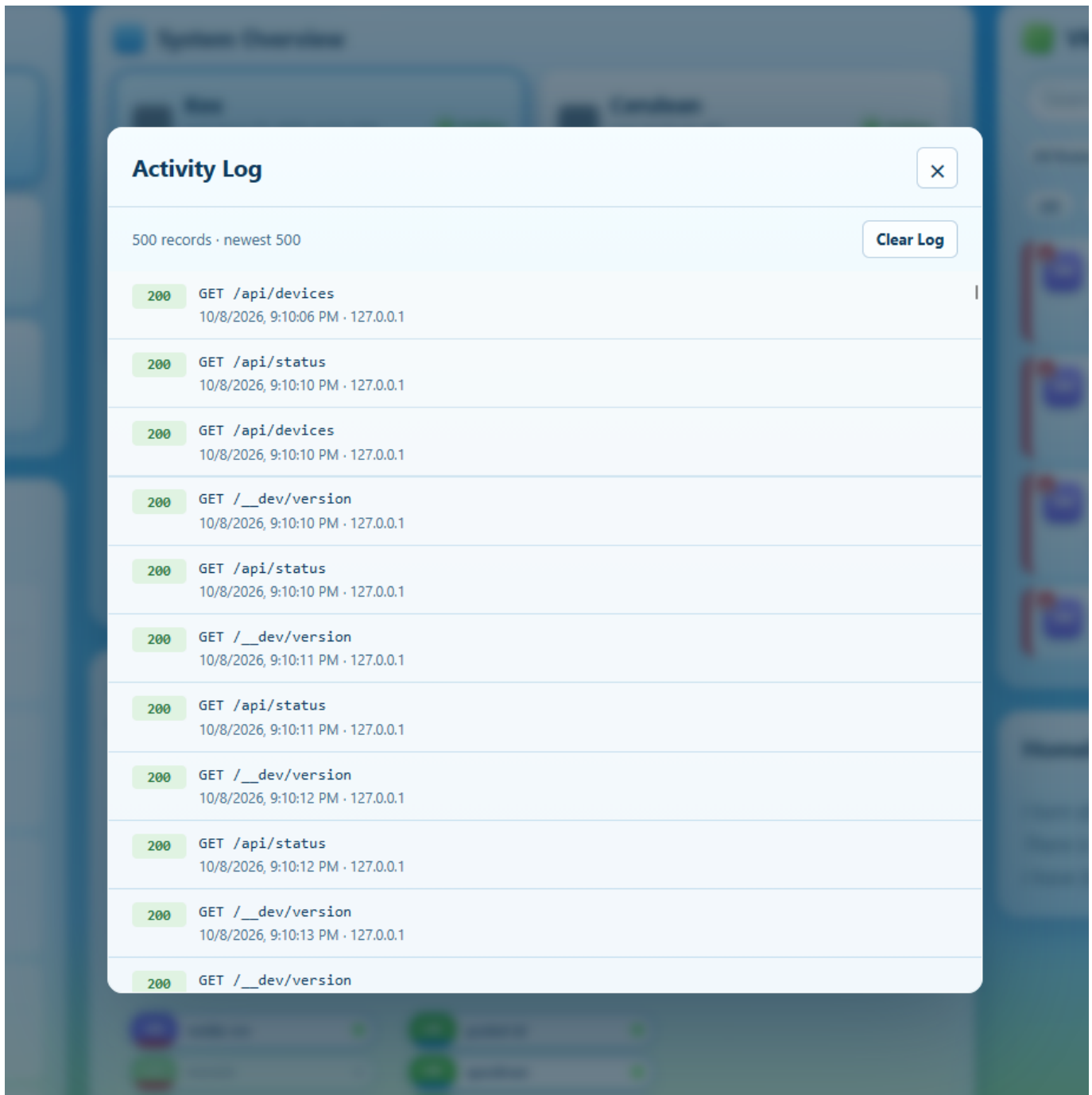


Figure 10: Activity log modal showing recent dashboard requests

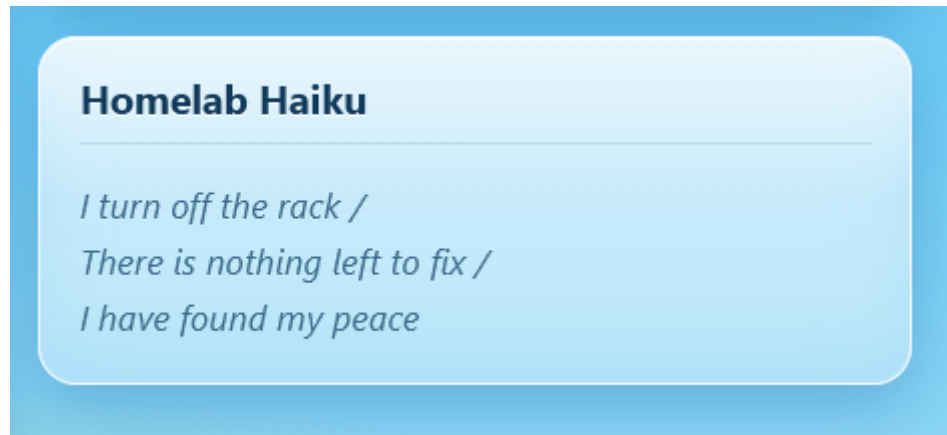


Figure 11: Homelab haiku panel



Figure 12: Dashboard background styling with the LAN-and-war cowsay message